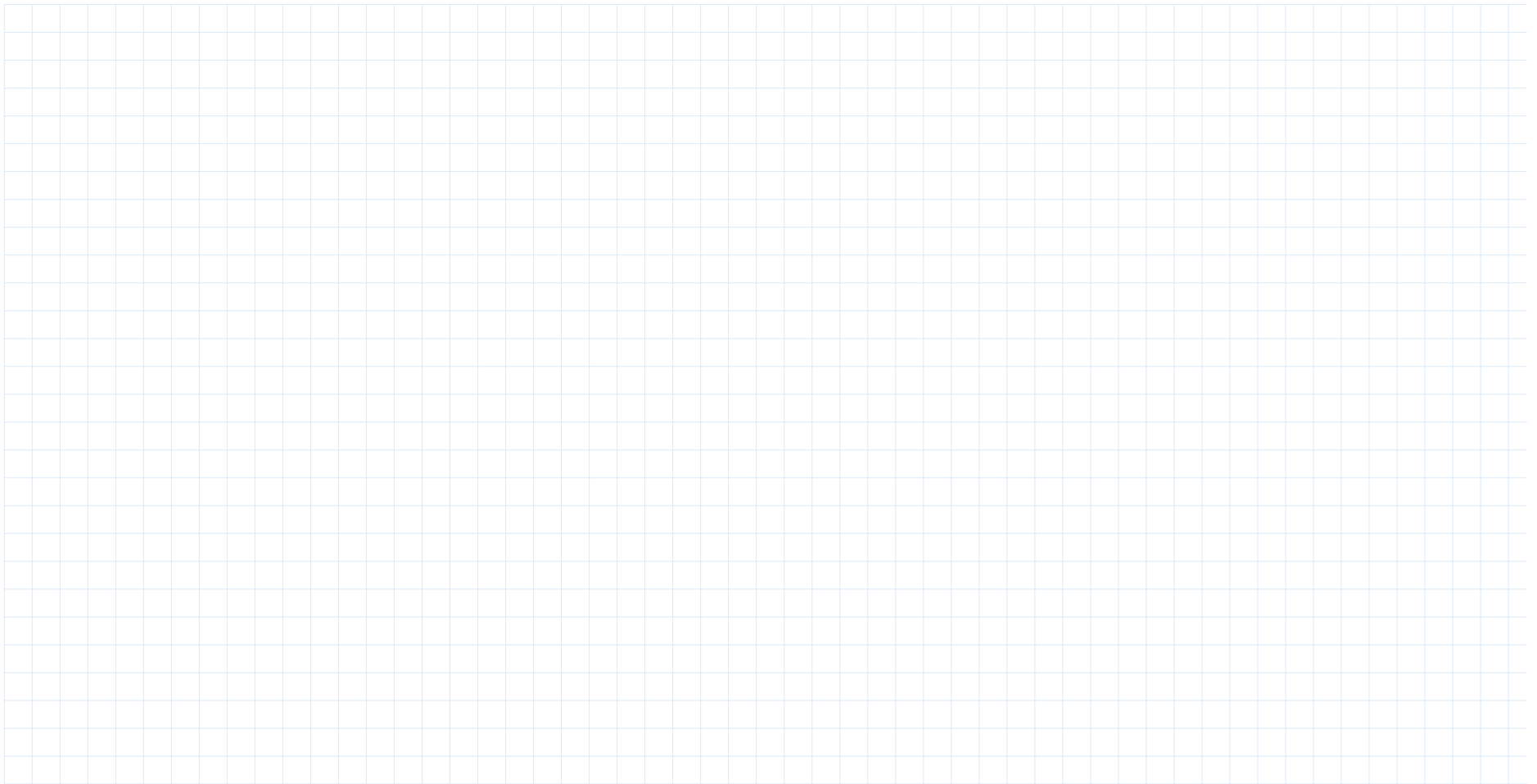


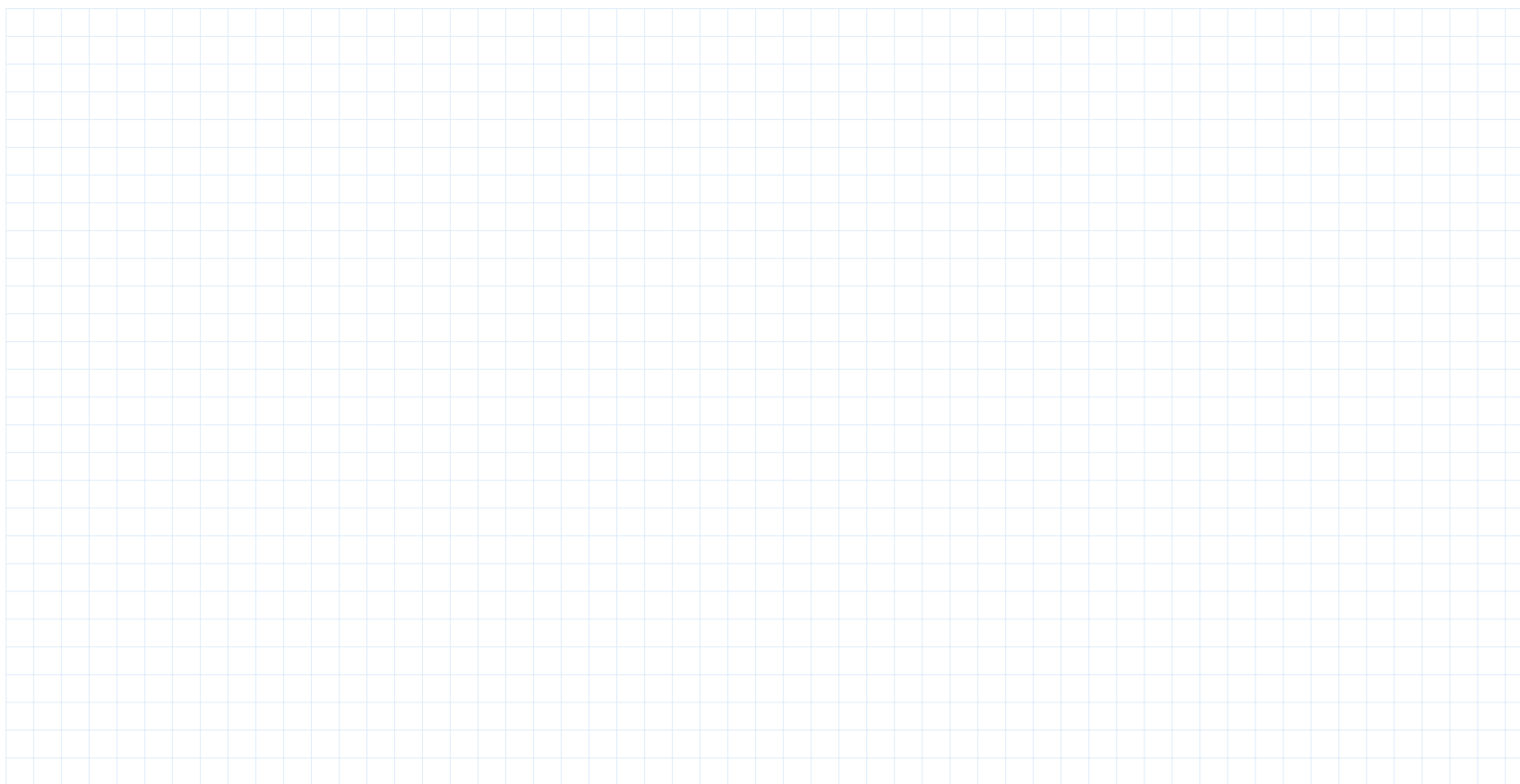
Solve a Trigonometric Equation

A trigonometric equation may have several solutions. The specified interval tells us which ones to keep. Solve on $0 \leq \theta < 2\pi$:

$$2 \cos \theta + 1 = 0.$$



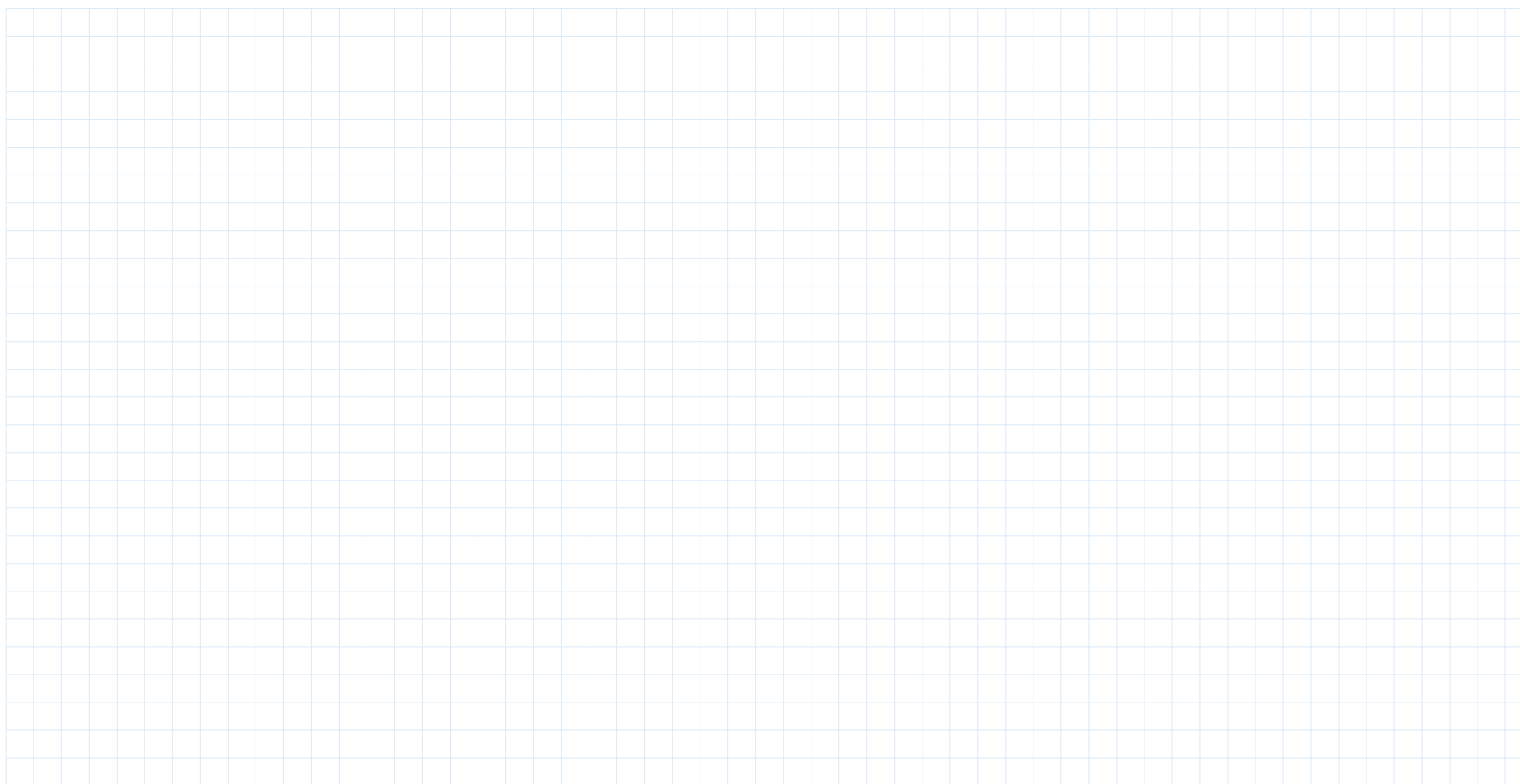
For all real solutions, add integer multiples of the function's period. Solve $\sin \theta = -\frac{\sqrt{2}}{2}$. **(a)** Give the general solution. **(b)** List the solutions in $[-2\pi, 2\pi)$.



Solve on $0 \leq \theta < 2\pi$:

$$\sin(2\theta) = \frac{\sqrt{3}}{2}.$$

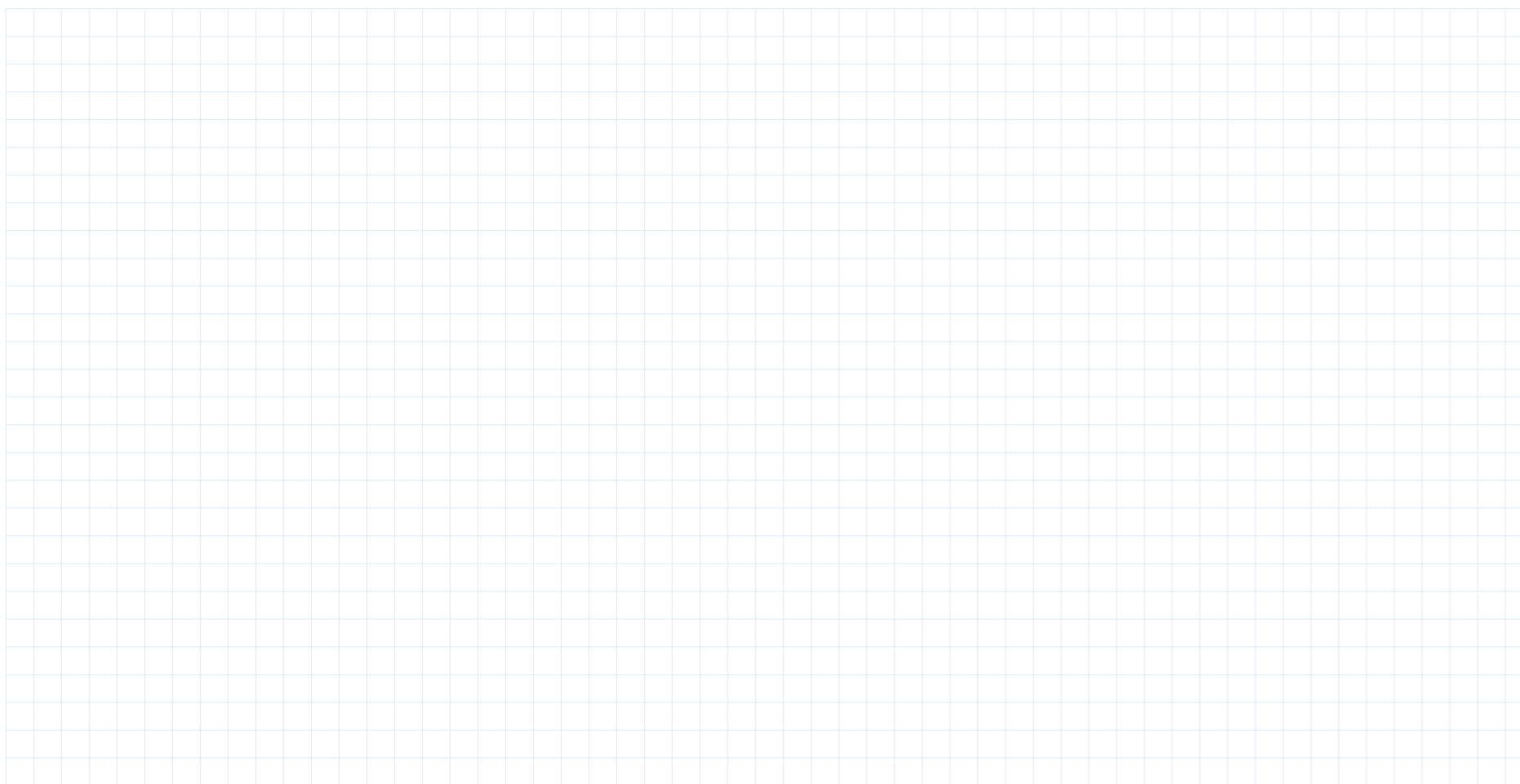
If $u = 2\theta$, first determine the interval for u .



Equations with Calculator Values

Section 6.3

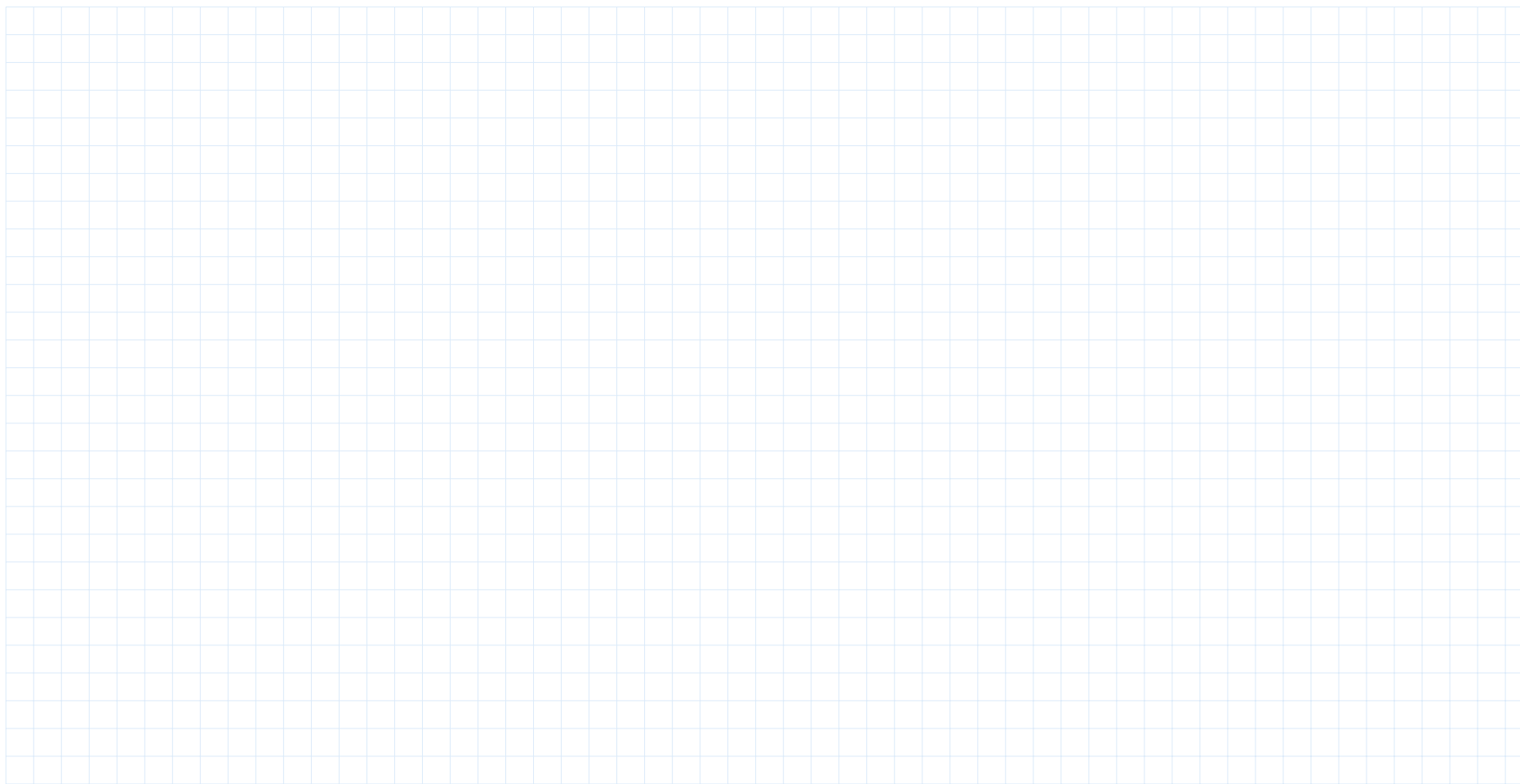
Solve $\tan \theta = -2$ on $0 \leq \theta < 2\pi$. Use radian mode and round each solution to two decimal places.



Solve on $0 \leq \theta < 2\pi$:

$$2 \sin^2 \theta + \sin \theta - 1 = 0.$$

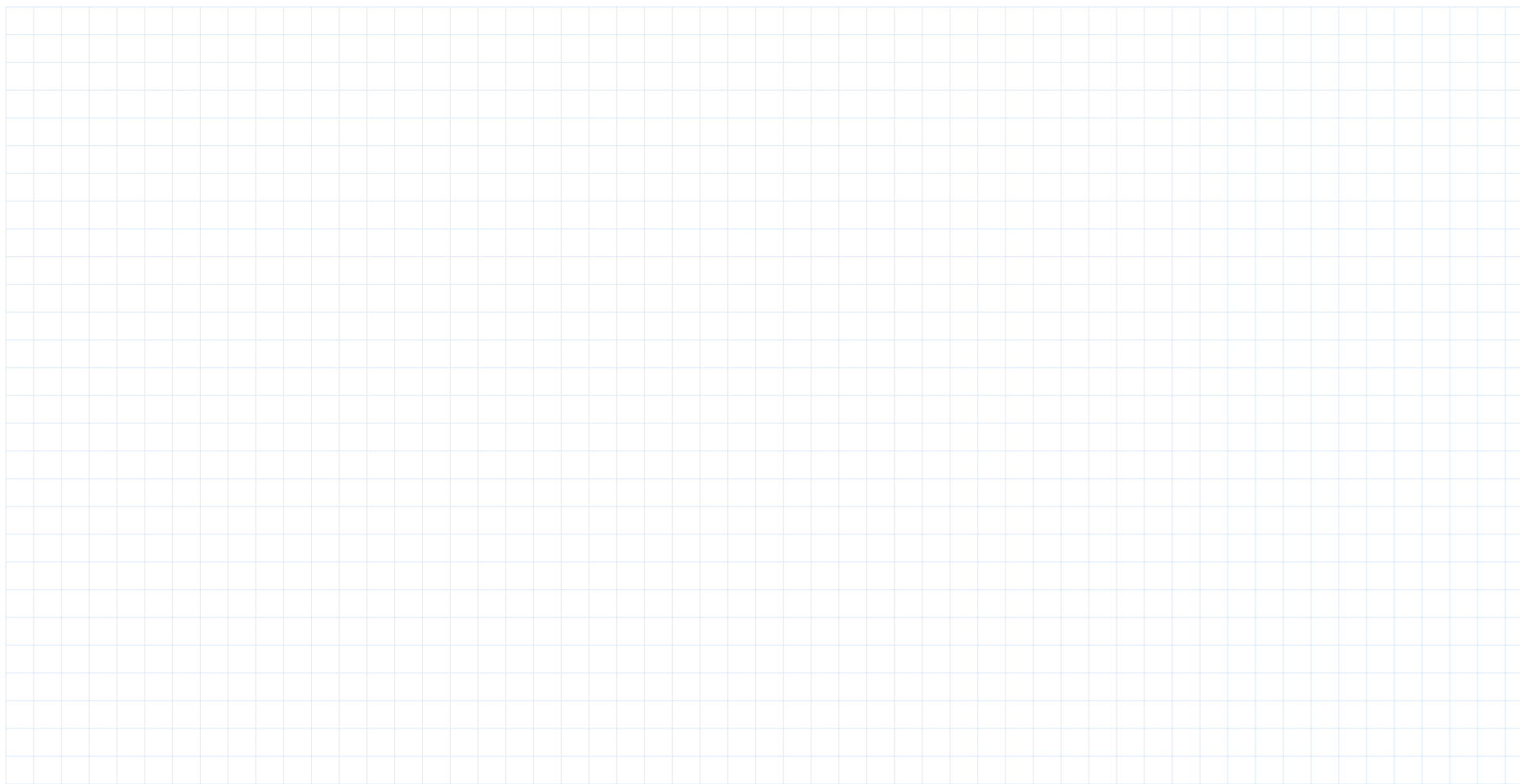
Treat $\sin \theta$ as one algebraic quantity when factoring.



Use an Identity to Solve

The identity $\sin^2 \theta + \cos^2 \theta = 1$ can rewrite an equation using one trig function. Solve on $0 \leq \theta < 2\pi$:

$$3 \cos^2 \theta = \sin^2 \theta.$$



A Graphical Solution

Use a graphing tool to solve on $0 \leq x < 2\pi$:

$$5 \sin x + x = 3.$$

Graph $y = 5 \sin x + x - 3$. Find all its zeros in the interval, to two decimal places.

